

Inkfinity: Augmenting Stylus-Based Handwriting with Contextual Awareness in AR

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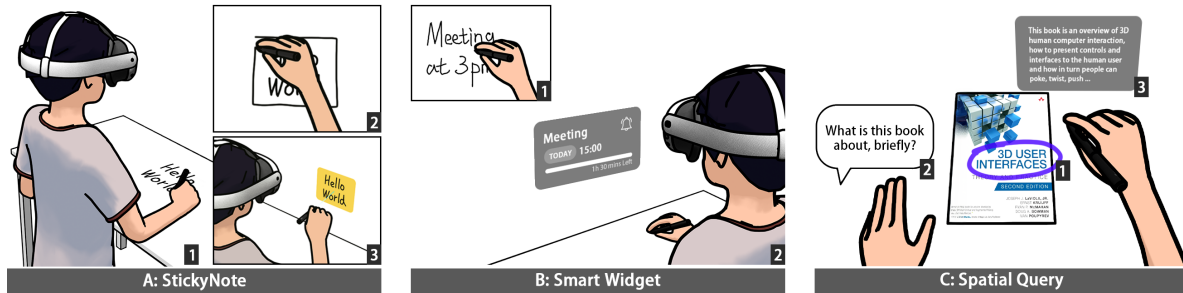


Figure 1: Core Interaction Workflows in Inkfinity. **(A) Sticky Note:** (1) User writes on a physical surface. (2) User encircles the handwriting to select it. (3) The selection becomes a manipulable 3D AR sticky note. **(B) Smart Widget:** (1) User creates a handwritten note (e.g., “Meeting at 3pm”). (2) A MLLM analyzes and converts the note into an interactive AR widget (e.g., Reminder). **(C) Spatial Query:** (1) User encircles a physical object (e.g., a book cover) in video see-through view. (2) User provides a voice prompt. (3) Contextual information is displayed as an AR overlay.

ABSTRACT

We present Inkfinity, an AR interaction paradigm bridging stylus-based spatial inking and actionable user intent through multimodal semantics. Using a spatially-tracked stylus, users write on physical surfaces and transform handwriting into interactive smart widgets via simple encircling gestures (e.g., converting “Meeting at 3pm” into a tappable reminder). Beyond digitization, Inkfinity enables contextual spatial queries: users encircle real-world objects to trigger AI-generated insights as AR overlays. By combining fluid stylus input with intelligent semantic understanding, this work demonstrates a seamless workflow for augmenting productivity in everyday environments.

Index Terms: Augmented Reality, Handwriting, Stylus Interaction, Large Language Models.

1 INTRODUCTION

Augmented Reality (AR) overlays digital information onto the physical world, creating new opportunities for how we work and learn. Yet many systems still treat users as consumers of pre-authored content, limiting agency in everyday productivity [10].

Ideally, AR’s ability to visualize data and provide contextual overlays should enable active and iterative workflows [4], but how to effectively support these processes has not yet been thoroughly explored. One major challenge lies in input methods. Current AR practices primarily rely on hand gestures, voice commands, and touch interfaces [2]. Yet few AR systems have explored the fluid and expressive input methods typically demanded by knowledge work, such as capturing ideas or organizing thoughts spatially. Another critical challenge is AR’s limited understanding of physical context. Traditional AR systems often rely on predefined markers or pattern recognition, constraining their ability to interpret and respond effectively to rich, unstructured real-world information.

Recent advances help address this gap: spatially-tracked styluses enable precise surface writing and mid-air gestures [7], and Multimodal Large Language Models (MLLMs) (e.g., GPT-4o) can interpret handwritten images and generate context-aware responses from the device view.

We present **Inkfinity**, an AR system that combines spatially-tracked stylus input with MLLM-powered interpretation. It allows users to transform handwriting on physical surfaces into interactive AR notes and query real-world objects with voice prompts. We contribute: (1) a stylus-based interaction paradigm that transforms spatial handwriting into semantically-aware smart widgets; and (2) a spatial query mechanism for retrieving contextual information about physical objects through MLLM analysis.

2 RELATED WORK

Handwriting with Digital Augmentation. Prior work has explored integrating physical handwriting with digital augmentation. Hybrid Stylus enables fluid cross-surface inking by mirroring handwritten strokes from paper to a tablet [3]. In immersive contexts, iVRNote lets VR users write on physical tablets for richer post-hoc review [5], and MaRginalia applies this concept to AR for in-lecture annotation [9]. Despite these advances, handwriting typically remains confined to planar surfaces and stored as static digital ink without semantic understanding, limiting transformation into actionable AR elements.

Spatial Ideation and Knowledge Work. Beyond planar capture, Phan et al. lift handwritten sticky notes into an AR brainstorming canvas via voice-triggered computer vision [8]. Surveys reveal that XR knowledge work research has predominantly focused on adapting keyboard and touch interactions to immersive environments [1], with less attention to expressive free-form input modalities.

Contextual Awareness in AR. Enhancing on-demand information access has moved beyond markers toward ad-hoc scene understanding. XR-Objects uses MLLMs to provide object-centric contextual menus for recognized physical items without pre-registration [6], illustrating how LLM-driven semantics can supply

immediate, context-sensitive feedback in AR.

3 SYSTEM

Inkfinity runs in Unity on Meta Quest 3 and uses a spatially-tracked stylus (MX Ink) [7] for three workflows that connect physical handwriting to semantic AR behaviors.

3.1 From Physical Notes to Interactive Widgets

Using the spatially-tracked stylus, users can write on any physical surface, such as a desk. To digitize, they encircle strokes with a bounding box—detected via heuristic-based shape recognition (closed loop with approximately straight edges). This converts the handwriting into a 3D AR sticky note, detachable from the surface and repositionable via stylus raycasting (select + drag).

To move beyond mere digitization and imbue these notes with intelligence, the handwritten content is captured as an image and sent to GPT-4o with a prompt. The prompt asks the model to recognize the text and classify intent into predefined categories (reminder, checklist, timer). If classified, the model returns a structured JSON. For instance, the time-specific entry “Meeting at 3pm” yields `{"type": "reminder", "title": "Meeting", "date": "Today", "time": "15:00"}`. This structured output is parsed, and Inkfinity replaces the original sticky note’s visuals with an interactive smart widget UI such as a Reminder; if the output is not in our predefined categories or parsing fails, it remains a plain note. Likewise, a multi-line list such as “Milk, Eggs, Bread” becomes a tickable checklist, whereas a temporal phrase like “Focus 25 min” immediately launches a countdown timer.

3.2 Instant Contextual Insights via Spatial Queries

Inkfinity adopts a screenshot-to-query metaphor for AR. With the spatially-tracked stylus, a user encircles any object or region in the video see-through view, for example, a paragraph in a printed book or an unfamiliar device. The user can then optionally issue a short voice query such as “summary” or “What is this”, or rely on a default prompt (e.g., “Describe the circled content”). The system captures the image inside the circle and sends it, together with the voice query (if provided) or a default prompt, to an MLLM. The model’s reply appears as a sticky note at the center of the circle, delivering context-aware information anchored to the selected content.

3.3 Spatial Stylus Interaction

A spatially-tracked stylus underpins Inkfinity’s workflow, supporting both precise surface handwriting and expressive mid-air gestures. Pressing and holding its side button gathers nearby AR notes into a compact stack that follows the pen tip. Users can carry this stack anywhere and release the button to trigger an auto-layout that fans the notes toward the viewer for clear viewing. If released over a recognized plane such as a wall or desk from the room scan, the notes pin to that surface, mirroring the familiar action of placing paper sticky notes on a whiteboard.

3.4 Implementation

Local. Inkfinity runs in Unity on Meta Quest 3 and uses a Logitech MX Ink spatially-tracked stylus for input; Unity handles gesture detection, image capture and cropping, widget management, rendering, and JSON validation/fallback.

MLLM. GPT-4o handles handwriting recognition, intent classification, and contextual query responses via API calls.

4 DISCUSSION & CONCLUSION

We presented Inkfinity, an AR system that combines spatially-tracked stylus input with MLLM-powered analysis to enable seamless physical-digital workflows. By reducing interaction friction

and extending users’ cognitive processes into the physical environment, Inkfinity demonstrates how expressive input modalities combined with semantic understanding can augment everyday productivity.

Informal trials with ten participants revealed promising feedback: users found the handwriting-to-widget conversion intuitive and described it as significantly reducing interaction friction. The mid-air encircling gesture was easy to learn. Widget creation latency is dominated by MLLM API calls (~1s), while local capture completes in ~100ms, which participants found acceptable for the added intelligence.

Limitations. MLLM-based handwriting recognition may occasionally produce inaccurate results. In 30 trials across five different handwriting styles with 5–12 words per note, we observed an 83% first-attempt recognition success rate. We also acknowledge that extended use of head-mounted displays for note-taking tasks may not be practical for everyday scenarios. While our current implementation relies on a spatially-tracked stylus, the interaction paradigm is transferable to hand gesture input (e.g., pinch-to-draw) and lightweight AR glasses, which we envision as more suitable form factors for ubiquitous deployment.

Future Work. We plan to diversify smart widget types, implement note editing functionality, and conduct comprehensive user evaluations on task efficiency and cognitive load.

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